

MTH209: Worksheet 2

Approximation by Truncation: Taylor Series

Many approximation methods in mathematics and statistics rely on expressing a complex object as a sum of simpler components and then truncating this expansion. A classic example of this is the Taylor series approximation.

Taylor Series: Function Approximation

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be smooth in a neighborhood of x_0 . The Taylor series expansion of f about x_0 is

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k.$$

The order- K Taylor approximation is obtained by truncation:

$$f_K(x) = \sum_{k=0}^K \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k.$$

The truncation order K controls the local accuracy of the approximation near x_0 and $f_K(x) \rightarrow f(x)$ pointwise as $K \rightarrow \infty$. Taylor series approximations are extremely popular in all of statistics and machine learning to build useful models and prove theorem.

Question: can a similar approximation be made for matrices?

Singular Value Decomposition

Let $A \in \mathbb{R}^{m \times n}$ be an arbitrary matrix with singular value decomposition

$$A = U \Sigma V^T.$$

The matrices U , Σ , and V have specific algebraic and geometric structures. The matrix $U \in \mathbb{R}^{m \times m}$ is orthogonal, satisfying

$$U^T U = U U^T = I_m.$$

Its column $u_1, \dots, u_m$ form an orthonormal basis of $\mathbb{R}^m$ and are called the **left singular vectors** of A since

$$A A^T u_i = \sigma_i^2 u_i.$$

The first $r = \text{rank}(A)$ columns of U correspond to the nonzero singular values and span the column space of A . The matrix $\Sigma \in \mathbb{R}^{m \times n}$ is diagonal, with nonnegative entries

$$\Sigma = \begin{bmatrix} \sigma_1 & & & & & \\ & \sigma_2 & & & & \\ & & \ddots & & & \\ & & & \sigma_r & & \\ & & & & 0 & \\ & & & & & \ddots \end{bmatrix},$$

where $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$. The number of nonzero singular values equals the rank of A . The matrix $V \in \mathbb{R}^{n \times n}$ is orthogonal, satisfying

$$V^T V = V V^T = I_n.$$

Its columns $v_1, \dots, v_n$ form an orthonormal basis of $\mathbb{R}^n$ and are called the **right singular vectors** of A since

$$A^T A v = \sigma_i^2 v.$$

The first r right singular vectors span the row space of A .

Reduced (Thin) SVD

In many applications, only the components corresponding to nonzero singular values are retained. The reduced SVD is

$$A = U_r \Sigma_r V_r^T,$$

where

- $U_r \in \mathbb{R}^{m \times r}$ has orthonormal columns,
- $\Sigma_r \in \mathbb{R}^{r \times r}$ is diagonal with positive entries,
- $V_r \in \mathbb{R}^{n \times r}$ has orthonormal columns.

This is the most commonly used version of SVD.

Low-rank Approximations

Equivalently, A admits the rank expansion

$$A = \sum_{i=1}^r \sigma_i u_i v_i^T,$$

where $r = \text{rank}(A)$, the singular values satisfy $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$, and $\{u_i\}$ and $\{v_i\}$ are orthonormal sets.

Using this sum representation, it is then possible to construct a low rank approximation of A . For $k < r$, The rank- k approximation of A is obtained by truncation:

$$A_k = \sum_{i=1}^k \sigma_i u_i v_i^T.$$

The rank k controls the global accuracy of the approximation. Both methods approximate by truncation, but Taylor series are inherently local, while low-rank matrix approximations are globally optimal.

Question: how to choose k ?

Answer: Note that each $u_i v_i^T$ is an outerproduct of unit vectors, and thus of equal “size”. It is the σ s that tell us the weight to be given to each matrix $u_i v_i^T$. The weights are ordered. Thus, k can be determined based on the quality of approximation. If a 95% approximation is sought, one can choose k to be the smallest index that satisfies

$$\frac{\sum_{i=1}^k \sigma_i}{\sum_{j=1}^r \sigma_j} > .95$$

That is, the smallest index that guarantees 95% of the accuracy.

Singular Value Decomposition for Positive Semidefinite Matrices

Let $A \in \mathbb{R}^{n \times n}$ be a symmetric positive semidefinite (PSD) matrix, that is, $A = A^T$ and $x^T A x \geq 0$ for all $x \in \mathbb{R}^n$. By the spectral theorem, A admits an eigenvalue decomposition

$$A = Q \Lambda Q^T,$$

where Q is orthogonal and

$$\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n), \quad \lambda_i \geq 0.$$

For a PSD matrix, the singular value decomposition takes the form

$$A = U \Sigma U^T,$$

where $U = Q$ and $\Sigma = \Lambda$. Thus, for PSD matrices: - singular values equal eigenvalues, - left and right singular vectors coincide, - the SVD and eigenvalue decomposition are identical.

Let $r = \text{rank}(A)$. Then A admits the expansion

$$A = \sum_{i=1}^r \lambda_i u_i u_i^T,$$

where u_i are orthonormal eigenvectors. Each term $\lambda_i u_i u_i^T$ is a rank-one PSD matrix. The rank- k approximation obtained by truncation is

$$A_k = \sum_{i=1}^k \lambda_i u_i u_i^T.$$

This is the best rank- k approximation to A in both the operator norm and Frobenius norm.

Problems

1. Consider the function $f(x) = e^{-2x^2}$ defined on all $\mathbb{R}$. Consider the Taylor series expansion of $f(x)$ around $x_0 = 1, 2, 3$. For each x_0 draw the approximation $f_K(x)$ of f for $K = 1, 2, 3, 4$ in the same plot.
2. Consider the following matrix

```
A <- matrix(1:12, nrow = 4, ncol = 3)
```

Using `svd(A)`, obtain the singular value decomposition. Verify that columns of U are left singular vectors and the columns of V are right singular vectors.

3. Consider any arbitrary matrix of size 500×100 . Plot $\|A - A_k\|_F$ as a function of k where $\|\cdot\|_F$ is the Frobenius norm. What do you observe? Is this is what you would expect?
4. Load the grayscale image of the dog, and convert the image into a matrix (make sure you set the working directory):

```
graydog <- load.image("graydog.jpg")

# plot and check dimensions
plot(graydog)
dim(graydog)

# store only the matrix of numbers
mat <- graydog[,1,1]
```

Using singular value decomposition and low rank approximations, obtain a value of k that captures upto 95% of the singular values.

5. Using `save.image()` command, save the low rank approximation in R. Compare the size of the original image and the size of this image. What do you observe?
6. You can do image compression to a colored image as well, by doing a low rank k approximation for each color channel (RGB) and then combining the channels. Implement this for the `dog` image in your repository.
 - i. Can you think of any disadvantage of this method?
 - ii. How else could we do low rank approximations in this case?